# Longest Continuous Subarray (medium)

## Problem Statement

Find the length of the longest contiguous subarray within an array of integers, where the absolute difference between any two elements in this subarray does not exceed a specified limit. The challenge lies in ensuring the subarray is continuous and the range (difference between the maximum and minimum element in the subarray) fits within the given threshold.

### Examples

1. **Example 1**:
   - **Input**: Array = [10, 1, 2, 4, 7], Limit = 5
   - **Expected Output**: 3
   - **Justification**: The longest subarray where the absolute difference between any two numbers is at most 5 is [1, 2, 4].
2. **Example 2**:
   - **Input**: Array = [4, 8, 5, 1, 7, 9], Limit = 3
   - **Expected Output**: 2
   - **Justification**: One possible longest subarray is [7, 9] where the difference between 7 and 9 is within the limit of 3.
3. **Example 3**:
   - **Input**: Array = [3, 3, 3, 3, 3], Limit = 0
   - **Expected Output**: 5
   - **Justification**: Since all elements are the same, the entire array is the longest subarray with differences of 0.

## Try it yourself

Try solving this question here:

Java

```java
import java.util.TreeMap;

public class Solution {
    // Function to find the longest subarray with absolute diff less than or equal to limit
    public  int longestSubarray(int[] nums, int limit) {
        int left = 0, maxLength = 0;
        // ToDo: Write Your Code Here.
        return maxLength;
    }
}

```

Test Case | Run Code Result | Stdout

```
[8,2,4,7]
4
```

Use Example Testcases | Run | Submit

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Solution: Longest Continuous Subarray

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